2 A force is a vector quantity.

(a) (i) State **two** features of a vector quantity.

1. .....

2. .....

[2]

(ii) State the names of **two** other quantities that are vectors.

1. .....

2. .....

[2]

(b) A student suspends a spring from a clamp stand and measures the length l_0 of the spring.

Fig. 2.1 shows the apparatus.

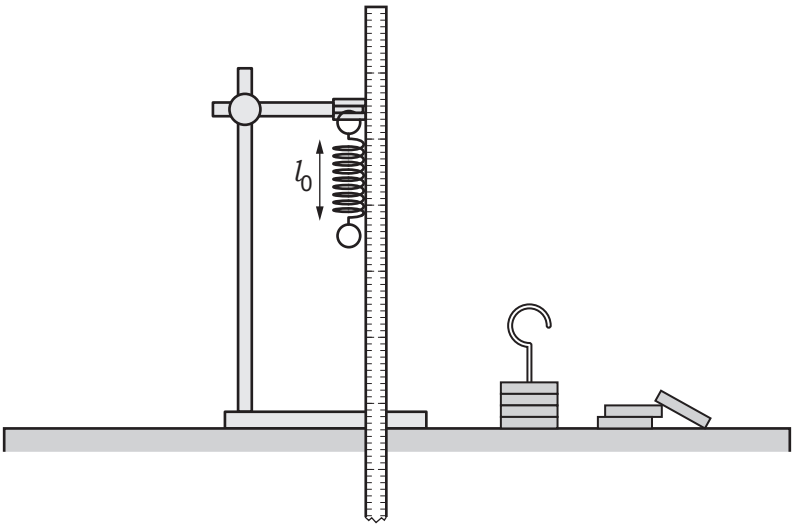


Fig. 2.1 (not to scale)

The student then suspends loads of different weights from the spring and measures the length of the spring for each load. He then plots a graph of the length of the spring against weight.

Fig. 2.2 is the graph that the student plots.

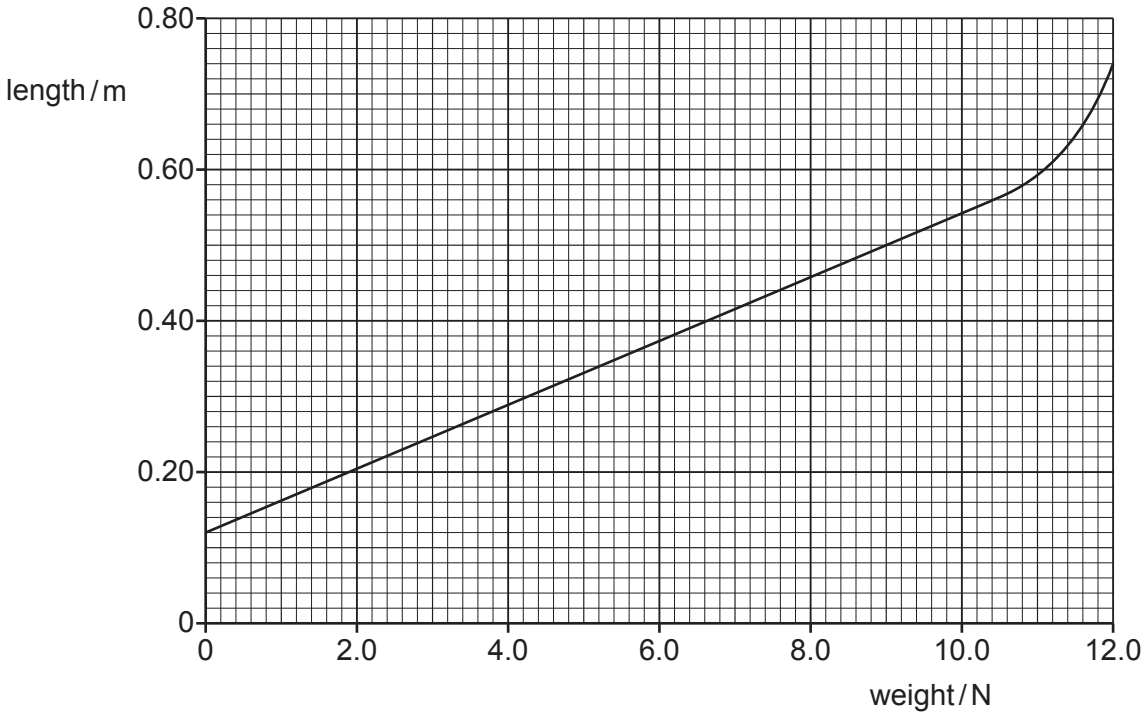


Fig. 2.2

(i) Using Fig. 2.2, determine the initial length l_0 of the spring.

l_0 = ... [1]

(ii) State what is meant by the limit of proportionality and, using Fig. 2.2, determine the weight of the load that causes this spring just to reach the limit of proportionality.

limit of proportionality .....

.....

.....

weight = ...

[2]

(iii) Using Fig. 2.2, determine the spring constant of this spring.

spring constant = ... [3]

[Total: 10]